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Insuresafe -AI-Powered Policy Recommendation and Customer Service Portal

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Project Guide
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Abstract

Insuresafe presents an AI-powered integrated insurance management system designed to enhance policy recommendation, identity verification, document validation, and claims processing. The platform features four core modules: a policy recommendation engine that utilizes user-specific factors such as age and income to suggest suitable insurance plans, a facial verification system leveraging convolutional neural networks (CNN) to securely authenticate users by comparing live selfies with stored photos during policy issuance, a document verification module that employs optical character recognition (OCR) and image classification techniques to extract and validate identity document details for regulatory compliance and fraud prevention, and an AI-assisted claims processing module that uses CNN-based analysis to validate uploaded claim documents, detect fraudulent activities, and automate approval workflows. Combining advanced machine learning algorithms with a user-friendly interface, the system aims to streamline insurance operations, reduce manual errors, enhance security, and improve user experience for both customers and insurance providers. This solution seeks to increase efficiency, trust, and transparency within the insurance ecosystem.

Sustainable Development Goals(SDG) mapped

SDG 9: Industry, Innovation, and Infrastructure→Leveraging AI and facial verification reflects innovation in financial services infrastructure.

Insuresafe uses smart AI tools like facial recognition and document checking to improve how insurance works. It helps verify people's identity, suggest the best insurance plans, and speed up claims in a safer and faster way. This makes the insurance system stronger, more reliable, and easier for everyone to use, which supports the goal of building better industries and infrastructure

Introduction

In the current insurance industry, customers face slow and complex processes for policy selection, identity verification, and claim approval. While existing solutions provide online access to policies and claims, they still require heavy manual verification, are prone to fraud, and lack personalized recommendations. These persisting issues highlight the need for a secure, automated, and user-friendly platform that combines advanced AI features to improve efficiency and trust.

Motivation

- Frequent cases of fake documents and fraudulent claims observed in the insurance market.
- Current verification systems rely heavily on manual checks, leading to delays and human errors.
- Growing demand for AI-driven, personalized insurance services that save time and reduce risk.

Objectives

- To integrate a policy recommendation engine to suggest suitable insurance policies based on user age or income.
- To implement facial verification using Deepface to compare a live selfie with the user's photo for secure identity validation during policy issuance.
- To develop a document verification module using OCR and image classification to extract and validate details from uploaded identity documents for compliance and fraud detection.
- To create an AI-assisted claims processing module using YOLO11 to validate uploaded documents/images, detect fraud, and automate claim approvals.

Literature Review

Sr. no	Title	Author(s)	Year	Methodology	Drawback
[1]	Deep Cross-Domain Insurance Recommendation System (DCDIR)	Ye Bi, Liqiang Song, Mengqiu Yao, Zhenyu Wu, Jianming Wang, Jing Xiao	2020	Graph-based insurance knowledge feature learning + GRU for user interest modeling + multi-layer perceptron for mapping across domains	Requires large cross-domain datasets; complexity in knowledge graph construction
[2]	Session-Based Insurance Recommendation Model	Simone Borg Bruun, Maria Maistro, Christina Lioma	2022	Recurrent neural network trained on user session sequences in insurance domain to recommend policies	Sparse interaction data; limited product variety; complex session modeling
[3]	Accurate and Interpretable ML for Transparent Pricing	Rohun Kshirsagar et al. (Lumiata team)	2020	Two-tier model (patient-level & employer-group-level) for prediction, outperforming actuarial models by 20%	High data requirement; restricted to employer-sponsored plans; complex implementation
[4]	Explainable AI-Enhanced Underwriting Automation	Hari Suresh Babu Gummadi	2025	XAI-enhanced underwriting using SHAP, ALE, counterfactuals, rule extraction; personalized recommendations	Heavy computational overhead; need for explainable frameworks; integration challenge

Sr. no	Title	Author(s)	Year	Methodology	Drawback
[5]	Machine Learning Recommender for Nigerian Health Insurance	Ayomide Owoyemi, Emmanuel Nnaemeka et al.	2023	Content-based recommender using KNN and cosine similarity for HMO selection based on location/prices	Limited to Nigerian context; doesn't address non-location factors
[6]	EfficientFace: An Efficient Deep Network with Feature Enhancement for Accurate Face Detection	Guangtao Wang, Jun Li, Zhijian Wu, Jianhua Xu, Jifeng Shen, Wankou Yang	2023	Lightweight CNN with feature fusion and attention for robust face detection	Struggles in highly occluded/low-resource cases; attention adds overhead
[7]	Human Face Recognition Based on CNN and Augmented Dataset	P. Lu, B. Song, L. Xu	2021	CNN with augmented training data to improve recognition accuracy	Relies on augmentations; limited generalization to real-world variations
[8]	Document Image Analysis Survey	V. V. Arlazarov, E. I. Andreeva, K. B. Bulatov, D. P. Nikolaev.	2022	Broad survey of document recognition techniques: OCR, layout, classification	No empirical baseline; broad coverage, not focused app
[9]	Document Classification via OCR	O. O. Petrova, B. I. Savelev, O. A. Slavin	2021	OCR+ML classifier pipeline for document categorization	OCR accuracy dependent on image quality; few real-world evaluations

Sr.no	Title	Author(s)	Year	Methodology	Drawback
[10]	Identity Document Verification Pipeline	Castelblanco et al.	2020	Deep learning ML pipeline: segmentation, localization, authenticity classification (~98% accuracy)	Tested on limited Colombian IDs; may not generalize
[11]	ConvNeXt V2 for Document Analysis	Xin, et al.	2023	Modern CNN-based architecture for classification of document images	Experimental; not evaluated on insurance document datasets
[12]	Auto Vehicle Claim Processing (CV & NLP)	Fernando et al.	2022	Hybrid solution: CNN for damage detection, OCR, plus NLP to extract claim data	Limited domain (vehicle); limited generalization; complex integration
[13]	Medical Insurance Data Analytics w/ DL	(Unir team)	2023	CNN pipelines for medical documentation and image analysis in health claims	Domain-specific; generalizability limited
[14]	AI Transformation in Healthcare Claims	ResearchGate authors	2025	Survey of NLP, Intelligent Document Processing, automation in claims	Descriptive; lacks experimental results
[15]	Medical Image Analysis Using DL	M. Li and co-authors	2023	Survey of deep learning techniques for medical images (transferable to claims)	General healthcare focus; needs adaptation for insurance workflows

Research Gap(Limitations of existing systems)

•**Manual verification delays** – Claims and policy approvals still rely on paperwork and repeated document submissions, slowing the process.

•**Ineffective chatbots** – Automated support often fails to resolve queries, leading to frustration and longer resolution times.

•**Old systems** – Insurance companies often use outdated systems where data is stored separately, making it slow to find and update customer information.

Problem Definition

- Difficulty in selecting suitable insurance policies due to lack of recommendations.
- Manual verification processes leading to errors and weak fraud detection.
- Slow claim settlement and outdated payment methods.
- Lack of AI-based tools like OCR, fraud detection, and chatbots.
- Complex and non-user-friendly insurance websites.
- Need for an intelligent platform to automate processes and improve efficiency.

Scope

•**Easy Policy Selection** – A simple interface with clear language and AI-driven recommendations to help users choose the best policy for their needs.

•**Secure Identity Verification** – Facial recognition and document scanning to confirm user identity quickly and safely.

•**Fast Document Verification** – AI-based OCR and image analysis to check the authenticity of identity documents without manual delays.

•**Smart Claim Processing** – AI tools to analyze claim images/documents, detect fraud, and approve genuine claims faster.

Technological Stack

Frontend Technologies :

- HTML5, CSS3, JavaScript
- Bootstrap 5.3.7

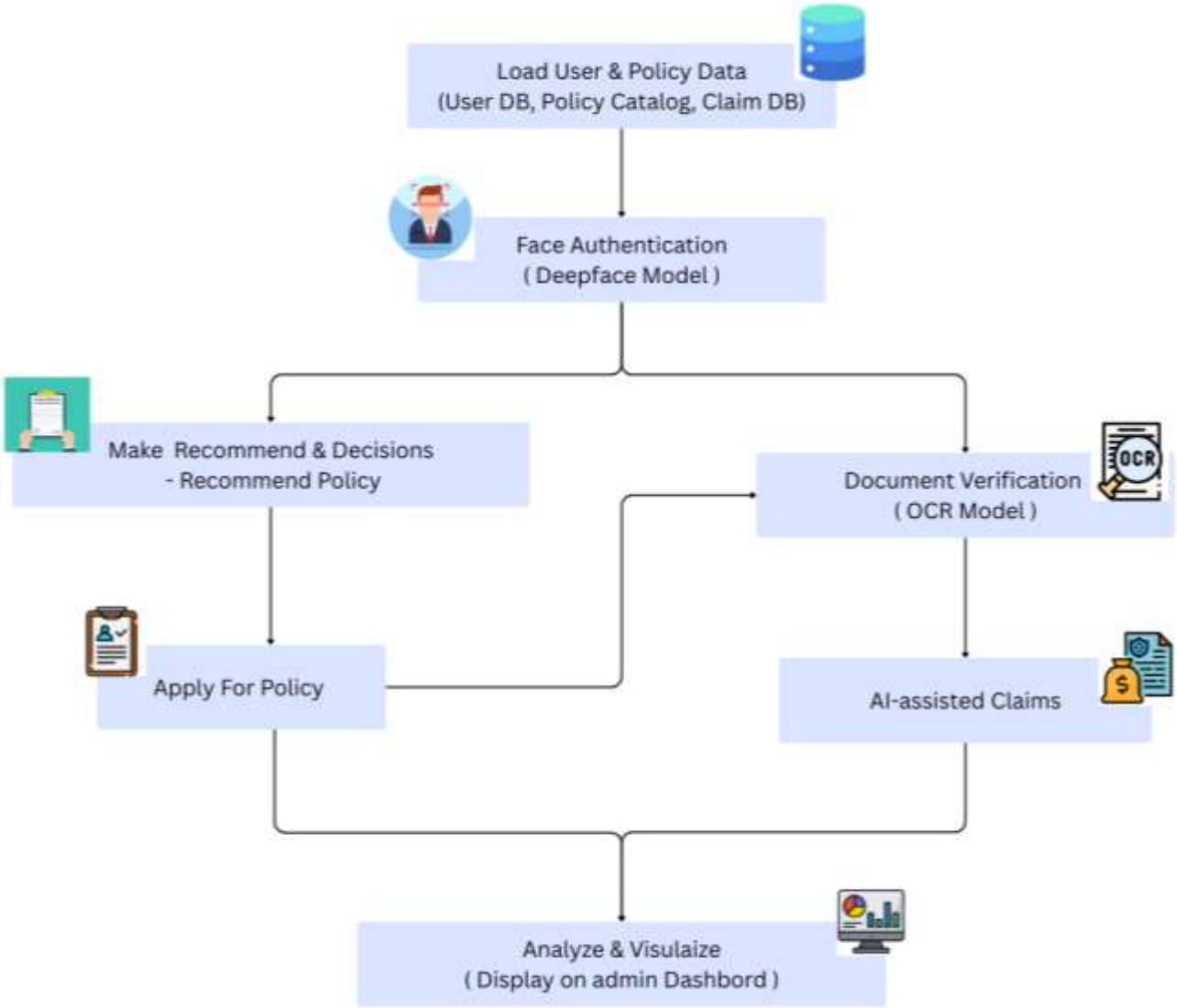
Backend Technologies :

- Python 3.13.1
- Flask 3.1.2
- MongoDB 1.46.11

Model Used:

- Deepface
- Optical Character Recognition (OCR)
- YOLO11

Proposed system architecture/Working/Module Diagram



Prototype Design Demonstration

Fig. 1.1 Landing Page

Prototype Design Demonstration

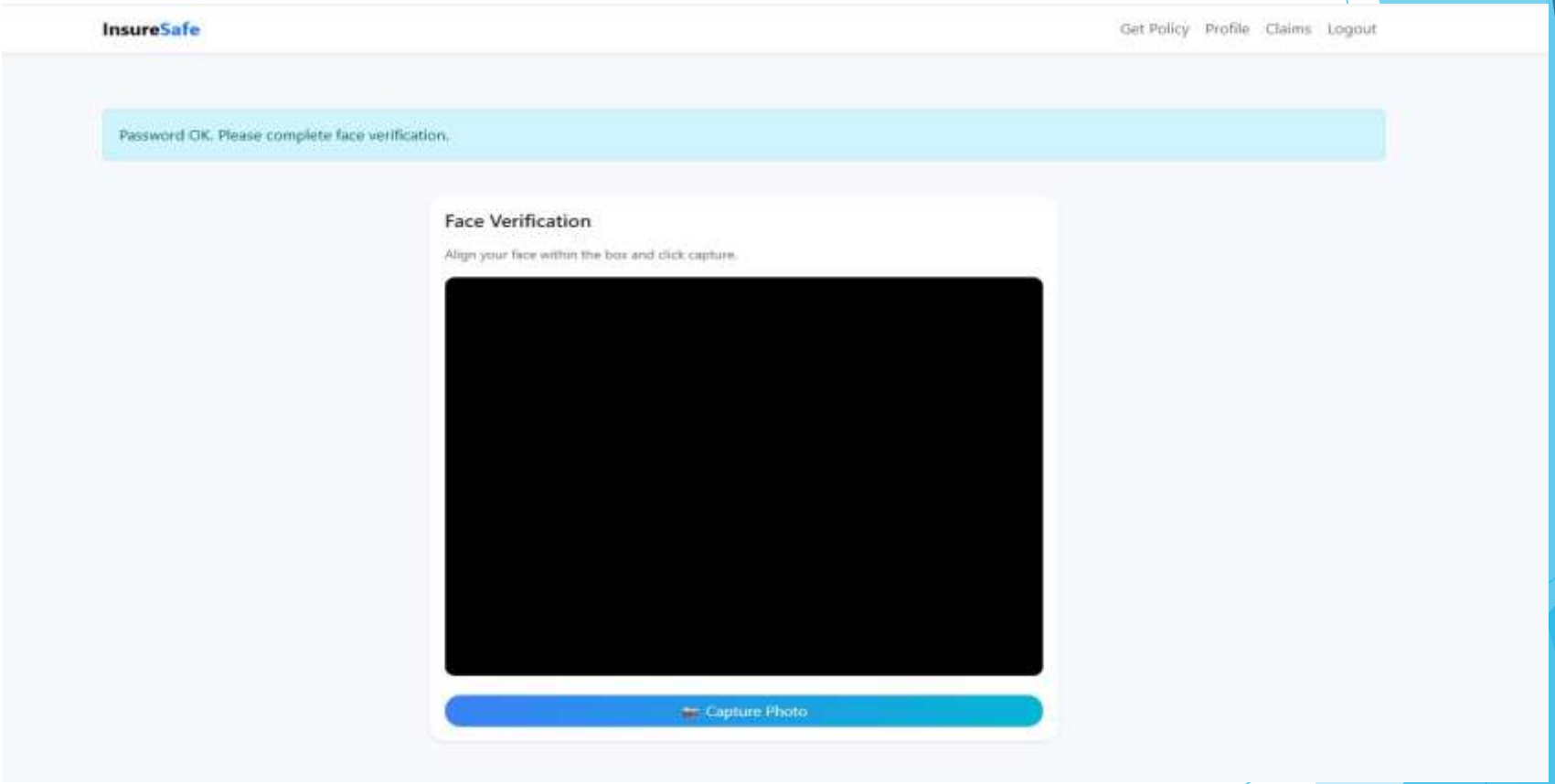


Fig. 1.2 Face Verification

Prototype Design Demonstration

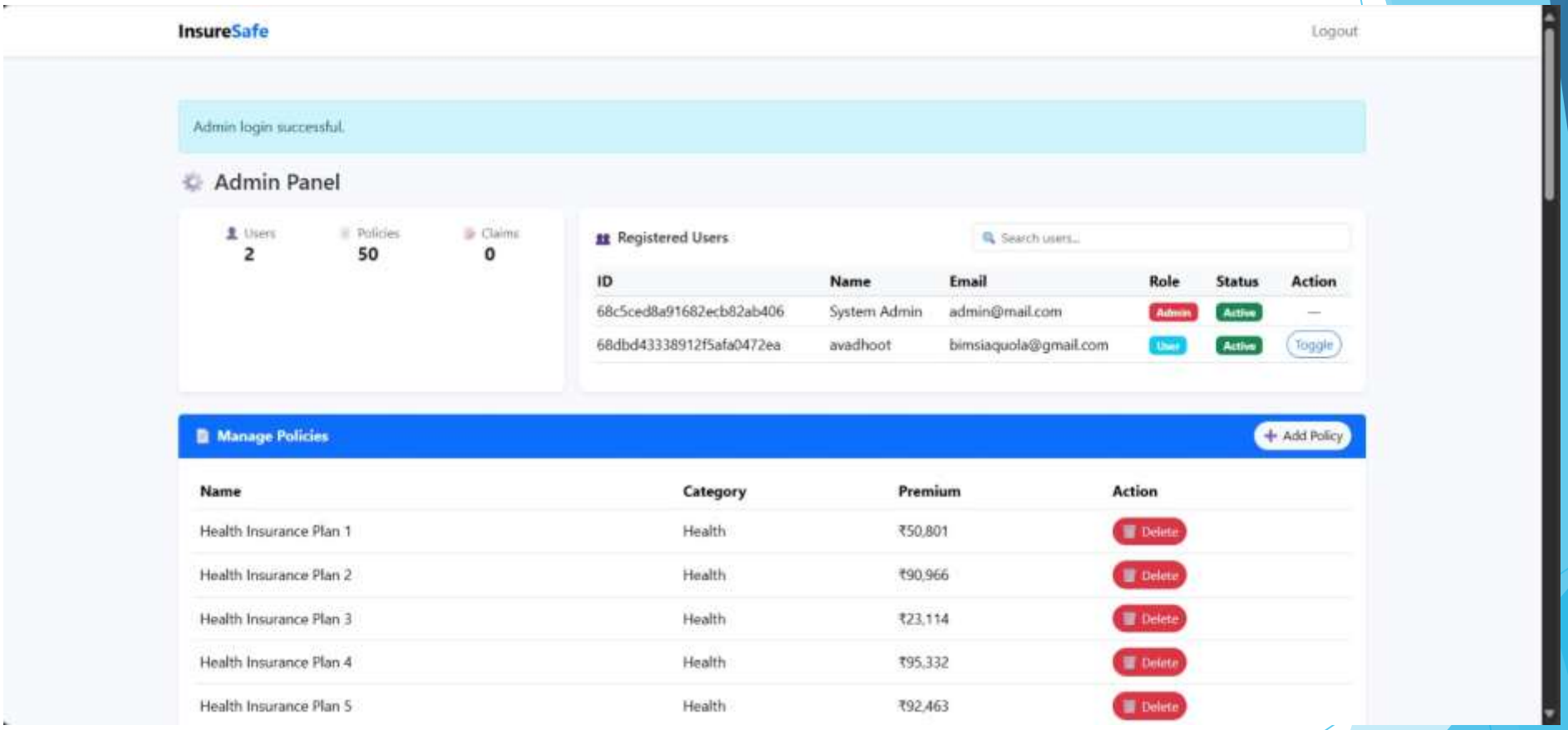


Fig. 1.3 Admin Panel

Prototype Design Demonstration

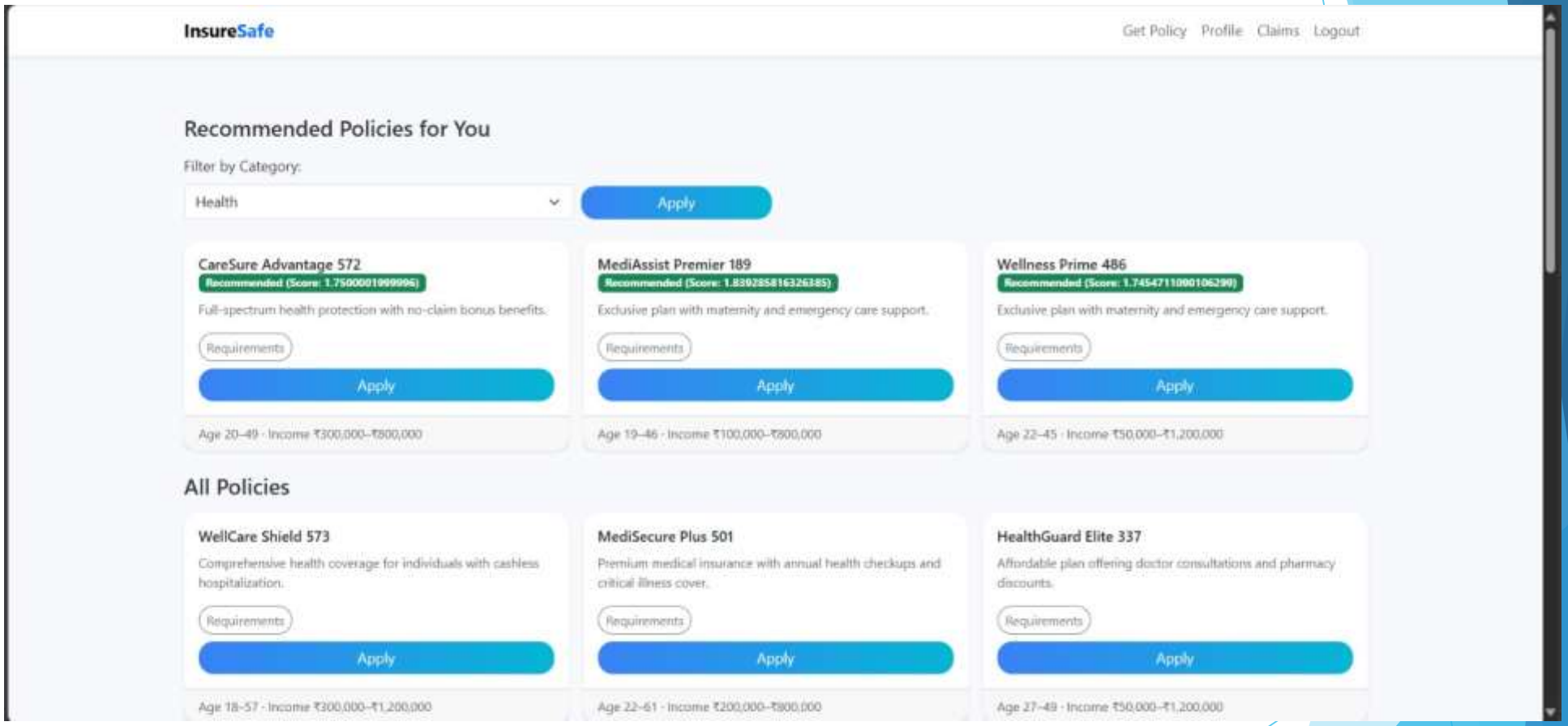


Fig. 1.4 Policies Page

Prototype Design Demonstration

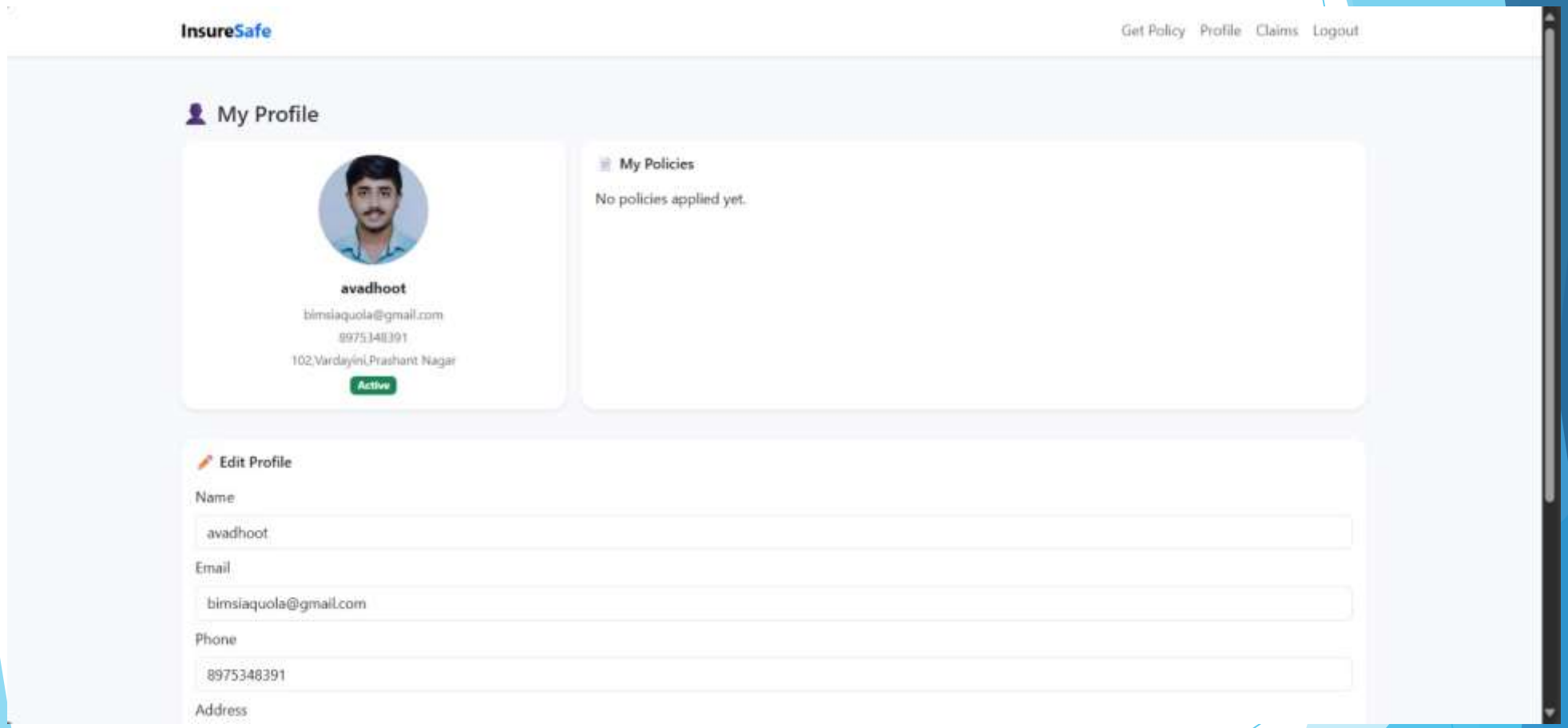


Fig. 1.5 Profile Page

Prototype Design Demonstration

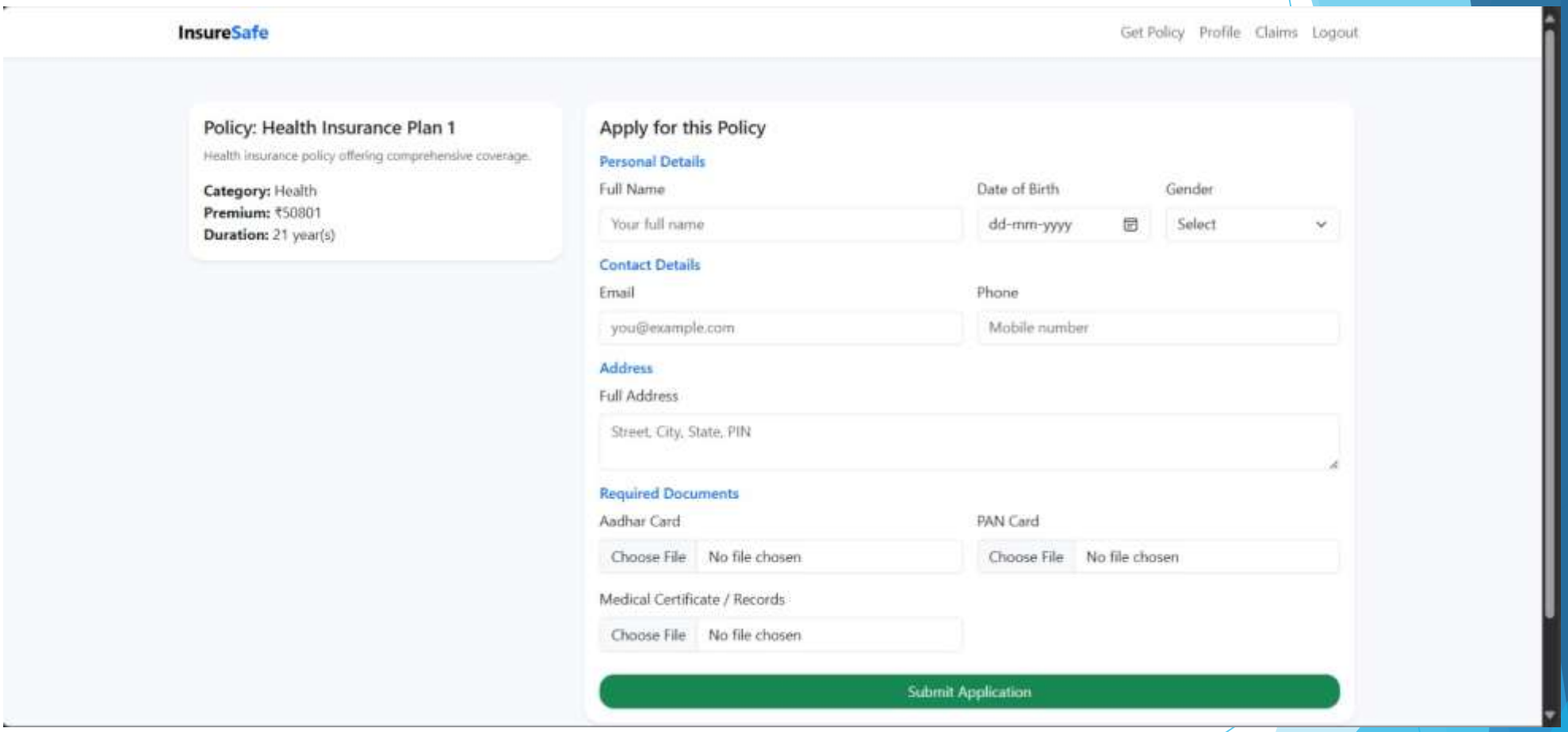


Fig. 1.6 Application For Policies

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Thank You...!!